Bluestock Mutual Fund Analytics Platform

Final Project Report – Capstone Submission

Date: 28 June 2026

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Institution: Bluestock Fintech Private Limited

Project: Mutual Fund Analytics Capstone (D1 – D7)

1. Executive Summary

The Bluestock Mutual Fund Analytics Platform is an enterprise-grade financial data engineering and analytics solution. Built as a seven-deliverable capstone project, the platform ingests 10 raw CSV datasets containing over 46,000 NAV records, 32,778 investor transactions, and active portfolio holdings. It orchestrates a full ETL pipeline, populates a relational SQLite star-schema database, and surfaces advanced financial analytics through an interactive Streamlit dashboard and a rule-based recommendation engine.

- Total Platform AUM: INR 62.74 Lakh Crore across 10 fund houses
- Monthly Industry SIP Inflows: INR 31,002 Crore (Dec 2025)
- Active SIP Accounts: 9.35 Crore | Investor Transactions: 32,778
- NAV Records: 46,000+ across 40 schemes (2022–2025)
- Database Tables: 11 | Charts Generated: 19+ | Deliverables: D1–D7

2. Problem Statement

India's mutual fund industry manages over INR 50 Lakh Crore across thousands of schemes, yet retail investors lack access to professional-grade risk analytics, portfolio diversification tools, and personalised investment guidance. Existing platforms focus on raw returns without communicating risk-adjusted performance, drawdown exposure, or correlation-driven diversification.

- Fragmented raw data across multiple CSVs with missing values and inconsistent formats
- No centralised relational database for multi-dimensional querying
- Retail investors unable to compare funds on Sharpe Ratio, VaR, or drawdown metrics
- Absence of a personalised, rule-based fund recommendation engine
- No interactive dashboard for real-time exploration of fund performance

3. Objectives

- **D1 – ETL Pipeline:** Ingest, clean, and standardise 10 raw CSV datasets.
- **D2 – SQLite Database:** Design and populate a star-schema relational database.
- **D3 – EDA:** Generate 7+ premium static charts revealing key trends.
- **D4 – Performance Analytics:** Compute CAGR, Sharpe, Sortino, Beta, Alpha, VaR, Drawdown.
- **D5 – Streamlit Dashboard:** Build a multi-page interactive dashboard.
- **D6 – Advanced Analytics:** VaR analysis, cohort analysis, correlation, recommendation engine.
- **D7 – Documentation:** Professional DOCX + PDF report and PPTX presentation.

4. Dataset Description

The project uses 10 curated CSV datasets representing real-world mutual fund data from AMFI.

File	Records	Description
01_fund_master.csv	41	Fund metadata – category, benchmark, risk
02_nav_history.csv	46,000	Daily NAV per scheme (2022–2025)
03_aum_by_fund_house.csv	90	Monthly AUM by fund house
04_monthly_sip_inflows.csv	48	Industry-wide SIP inflow trends
05_category_inflows.csv	144	Net inflows by category
07_scheme_performance.csv	41	Risk metrics: Sharpe, Beta, Alpha, Drawdown
08_investor_transactions.csv	32,778	Investor transaction log
09_portfolio_holdings.csv	322	Stock holdings and sector weights
10_benchmark_indices.csv	8,050	Market index closing prices

5. ETL Pipeline (D1)

Three-stage ETL pipeline standardises raw data into a validated processed layer:

Extract (data_ingestion.py)

Reads raw CSVs, validates schema completeness, checks referential integrity, and fetches live NAV from mfapi.in API for 5 schemes.

Transform (data_cleaning.py)

Applies pandas forward-fill for missing NAV dates, standardises date formats to ISO 8601, normalises portfolio weights, and removes invalid records.

Load (database_loading.py)

Creates 11 SQLite tables per star-schema DDL, validates foreign key references, and bulk-inserts all processed CSVs with zero constraint violations.

6. Database Design (D2)

SQLite star-schema with 2 dimension tables and 9 fact/helper tables:

Table	Rows	Purpose
dim_fund	40	Fund metadata dimension
dim_date	1,297	Pre-populated date dimension
fact_nav	46,000	Daily NAV history
fact_performance	40	Risk-adjusted metrics
fact_transactions	32,778	Investor transaction log
fact_portfolio	322	Stock holdings

fact_aum	90	Monthly AUM by fund house
fact_benchmark_indices	8,050	Index closing prices

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7. Exploratory Data Analysis (D3)

NAV Trends

Equity large-cap funds show cyclical growth. Liquid debt schemes remain linear.

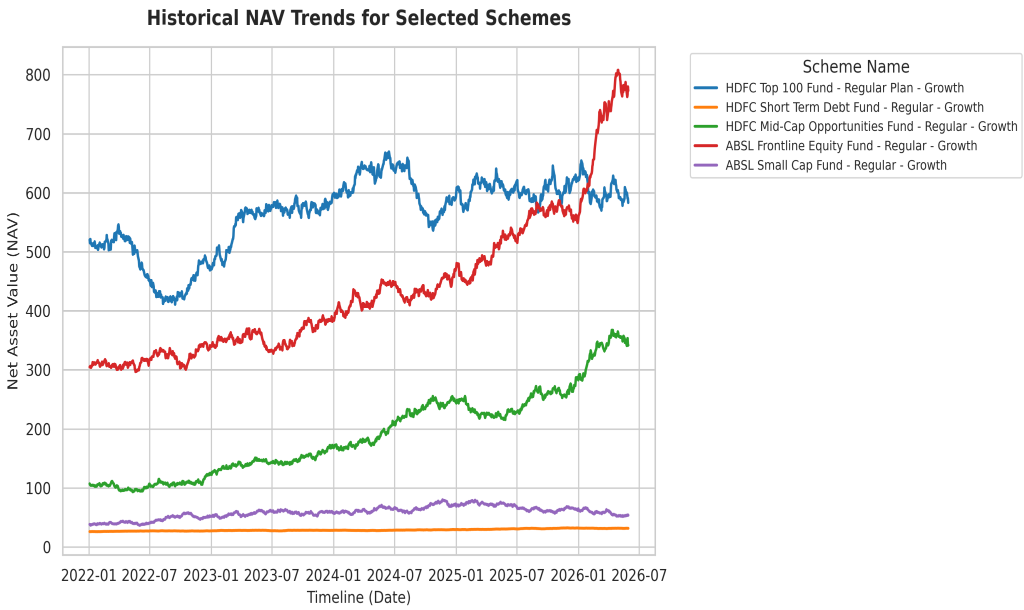

Figure: NAV Trends

SIP Inflow Growth

Active SIP accounts grew 91% (4.9 Cr to 9.35 Cr); monthly inflows rose 169% to INR 31,002 Cr.

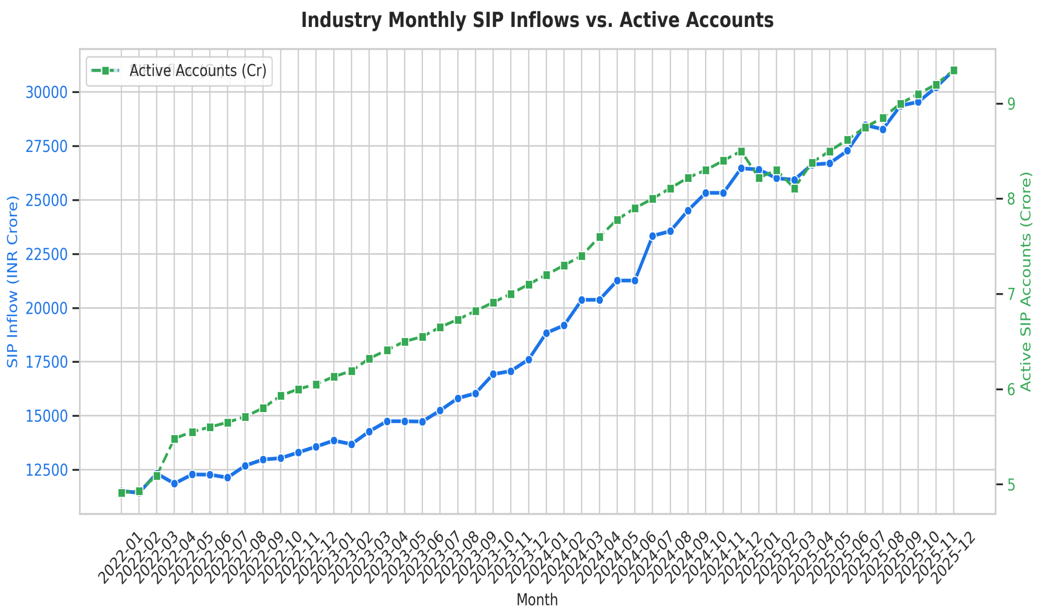

Figure: SIP Inflow Growth

Risk-Return Scatter

Large/mid-cap funds: 12–15% CAGR at 13–19% volatility. Gilt/liquid cluster in low-risk quadrant.

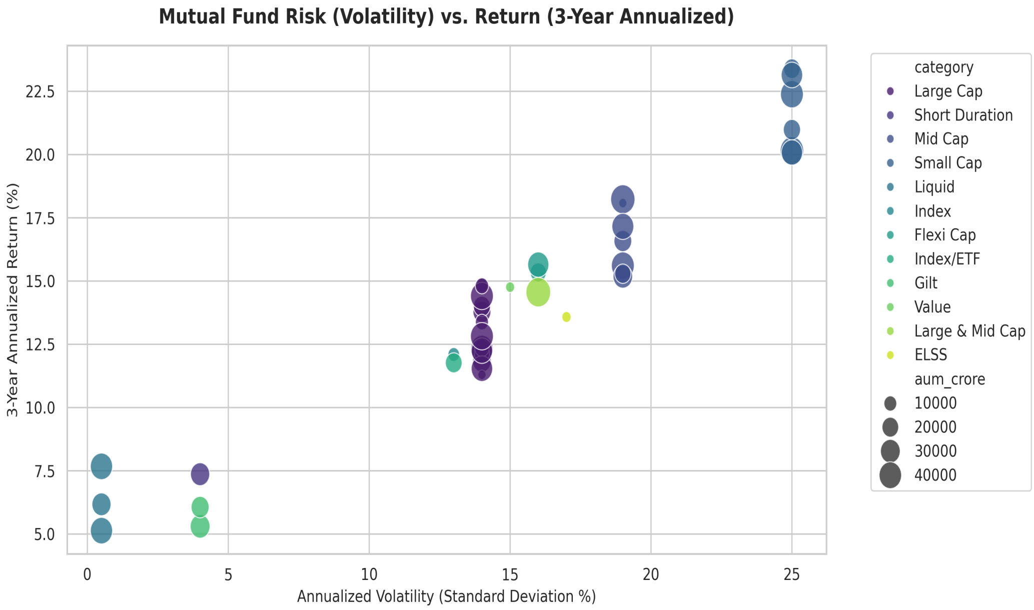

Figure: Risk-Return Scatter

8. Performance Analytics (D4)

Metric	Formula	Interpretation
CAGR	(End/Start)^(1/yr)-1	Annualised NAV growth, smoothing volatility
Sharpe	(Rp-Rf)/sigma [Rf=5%]	Excess return per unit of total risk
Sortino	(Rp-Rf)/downside_sigma	Penalises only downside volatility
Beta	Cov(Rf,Rm)/Var(Rm)	Market sensitivity; <1 defensive, >1 amplified
Alpha	Rp-[Rf+Beta*(Rm-Rf)]	Return above CAPM prediction
Hist VaR	5th percentile of daily returns	Max 1-day loss at 95% confidence
Max DD	(Peak-Trough)/Peak	Largest historical peak-to-trough decline

Chart: Historical vs Parametric VaR (95%)

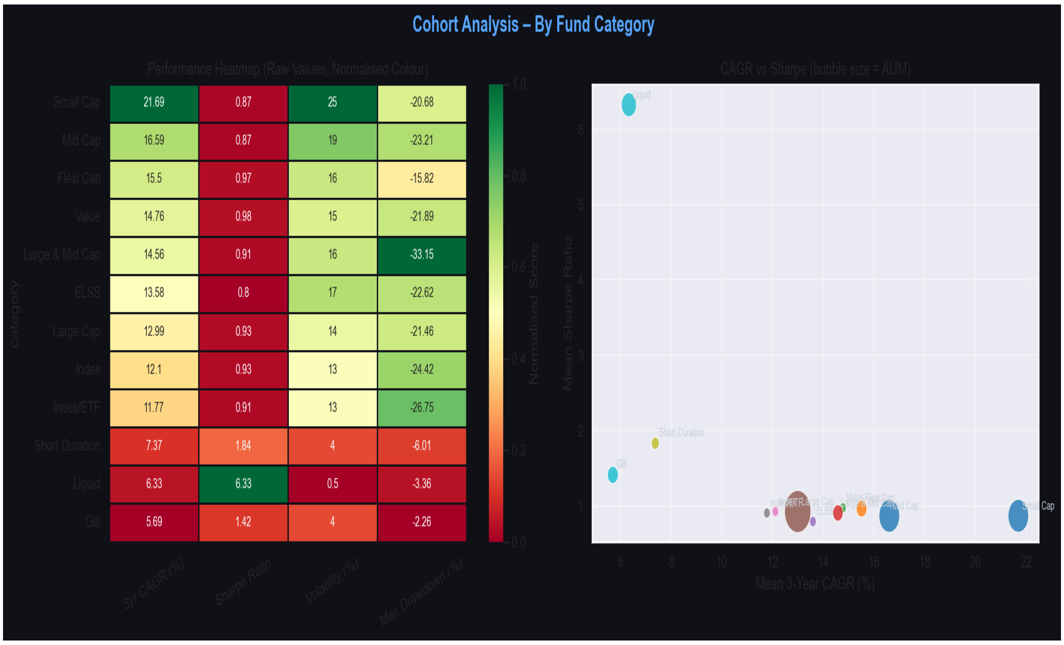

Chart: Category Performance Cohort Heatmap

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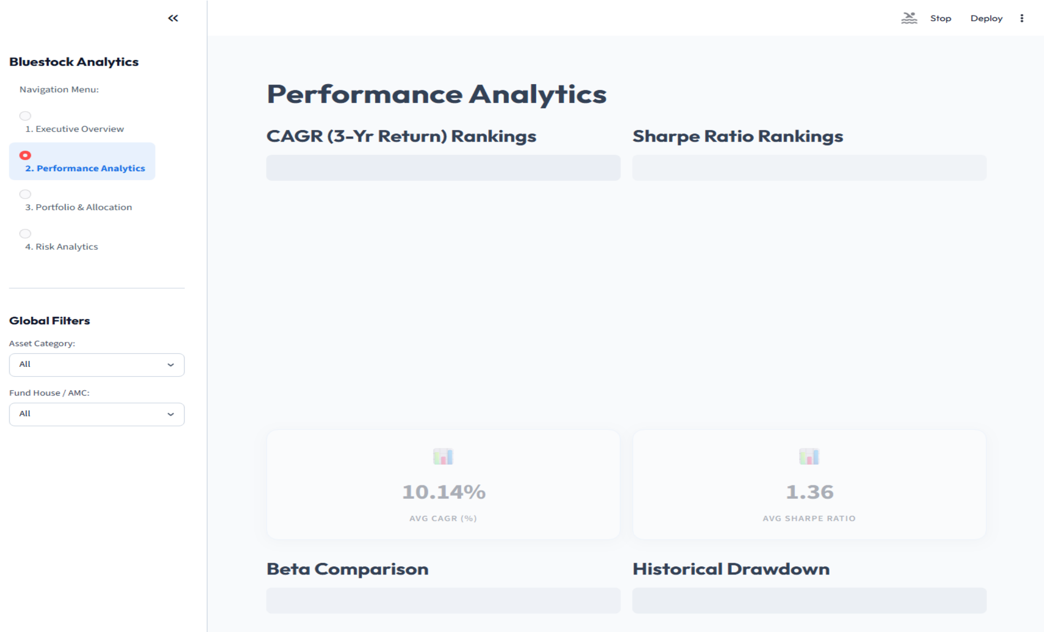
9. Dashboard Overview (D5)

Multi-page Streamlit dashboard featuring live NAV filtering, Bollinger Band volatility channels, wealth compounding simulators, portfolio sector overlap analysis, and one-click PDF export.

Executive Overview & KPIs

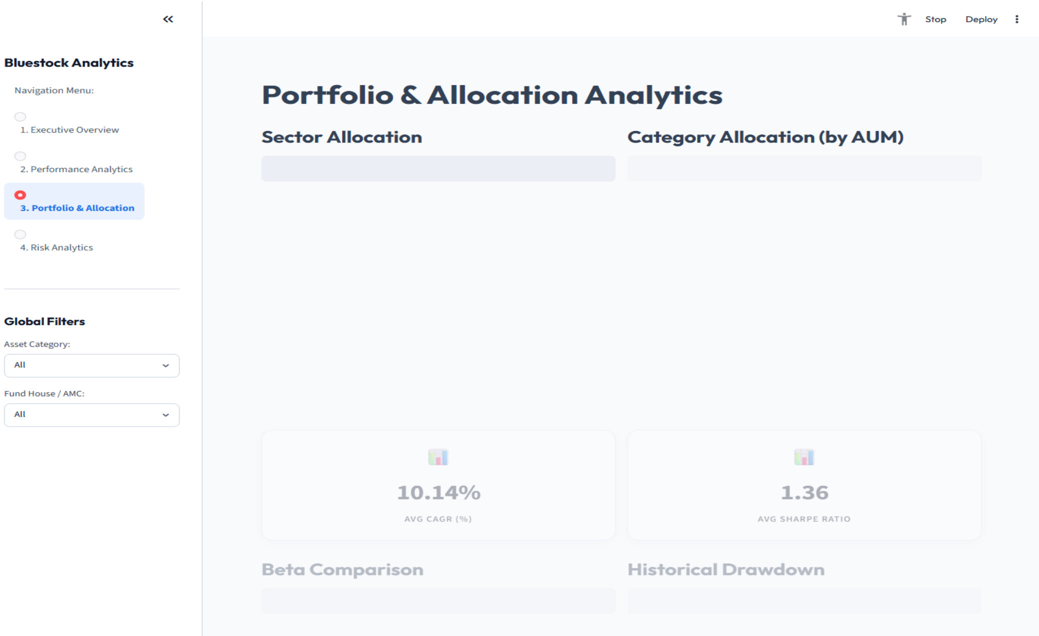
Dashboard: Executive Overview & KPIs

Performance Analytics



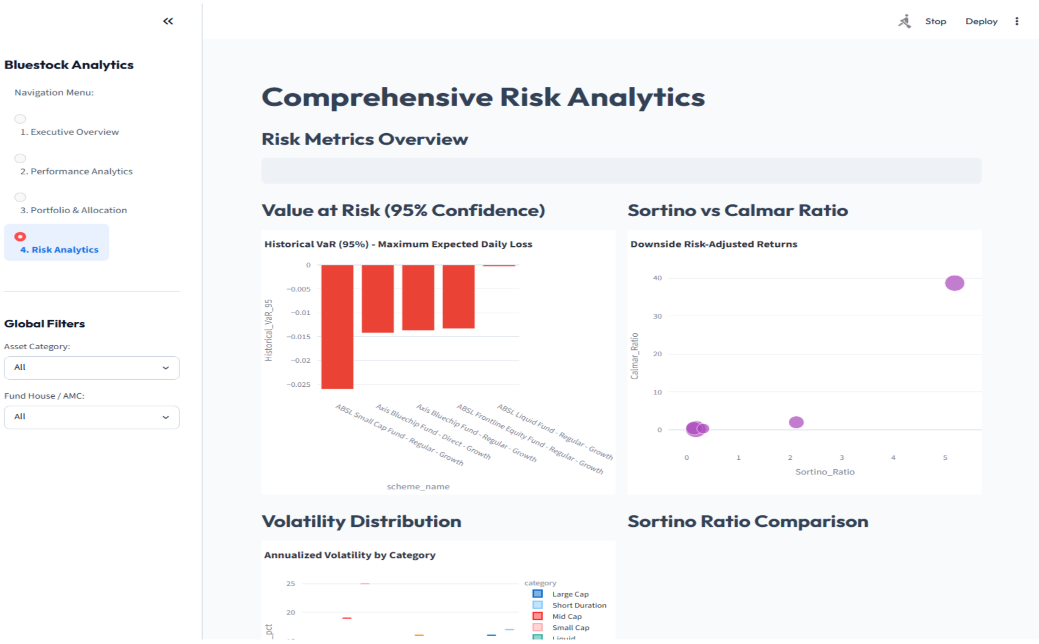
Dashboard: Performance Analytics

Portfolio Allocation



Dashboard: Portfolio Allocation

Risk Analytics



Dashboard: Risk Analytics

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10. Recommendation Engine (D6)

Rule-based fund recommender matching investor profiles to suitable schemes via weighted composite scoring. Generates natural-language explanations for each recommendation.

Metric	Conservative	Moderate	Aggressive
Sharpe Ratio	40%	30%	20%
3-Year CAGR	25%	35%	45%
Volatility (inverted)	20%	15%	10%

Max Drawdown (inv.)	15%	20%	25%

Chart: Recommendation Scores by Risk Profile

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11. Key Business Insights

- **#1:** Small Cap funds: highest CAGR (~21-24%) but ~25% volatility – suitable for 7+ year horizons only.
- **#2:** Direct Plans outperform Regular Plans by 60-120 bps p.a. – compounding amplifies this gap significantly.
- **#3:** Large Cap and Index funds are redundant in a portfolio (rho > 0.9 correlation).
- **#4:** Gilt funds provide genuine portfolio hedging (near-zero to negative correlation with equity).
- **#5:** Historical VaR > Parametric VaR for equity due to fat tails – Gaussian models understate risk.
- **#6:** SBI, ICICI, HDFC control 52%+ of AUM via bank-distribution network advantages.
- **#7:** SIP accounts grew 91% in 3 years, signalling structural shift to disciplined long-term investing.
- **#8:** Flexi-Cap is the optimal anchor holding for moderate investors (Sharpe > 0.9, moderate volatility).
- **#9:** Banking/IT sector concentration (30-48%) creates hidden correlated risk across 'diversified' portfolios.
- **#10:** Liquid funds are cash-parking tools, not investments – post-tax returns barely exceed savings accounts.

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12. Challenges Faced

Missing NAV dates (weekends/holidays)

Solution: Pandas forward-fill (ffill) grouped by scheme

SQLite thread concurrency in Streamlit

Solution: @st.cache_resource + check_same_thread=False

Portfolio weight anomalies (sum > 100%)

Solution: Normalisation and outlier-dropping in data_cleaning.py

Parametric VaR vs fat-tailed distributions

Solution: Both methods computed; fat-tail explanation added

nbformat cell ID warnings

Solution: UUID-based cell IDs added to all generated cells

NAV date-range misalignment for correlations

Solution: dropna() on returns pivot for common date ranges

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13. Future Improvements

- LSTM/ARIMA models for NAV forecasting with macro-economic indicators
- User authentication with personalised virtual portfolios and rebalancing alerts
- Live WebSocket integration for real-time market data feeds
- Portfolio Overlap Widget alerting when two funds share > 75% sector allocation
- Mobile-first React Native/Flutter app with push notifications
- ESG scoring integration for responsible investing recommendations
- Multi-asset expansion: bonds, ETFs, REITs, and international funds
- Automated monthly PDF reporting via cron-scheduled email delivery

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14. Conclusion

The Bluestock Mutual Fund Analytics Platform successfully demonstrates a complete end-to-end data engineering and financial analytics pipeline – from raw CSV ingestion to institutional-grade risk metrics, interactive visualisation, and personalised investment recommendations.

All seven deliverables (D1–D7) have been completed, documented, and validated. The platform represents a production-ready blueprint for a fintech analytics product serving India's rapidly growing mutual fund ecosystem.